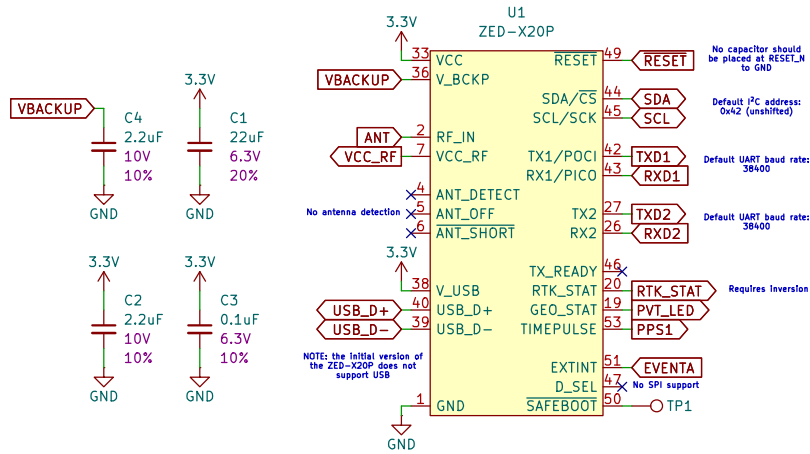
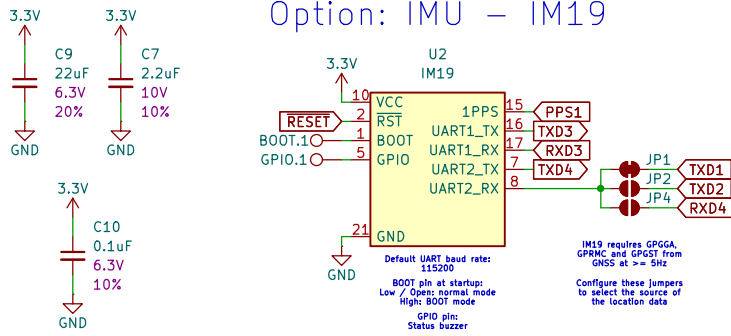


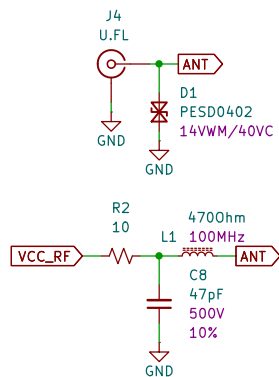
GNSS – ZED-X20P



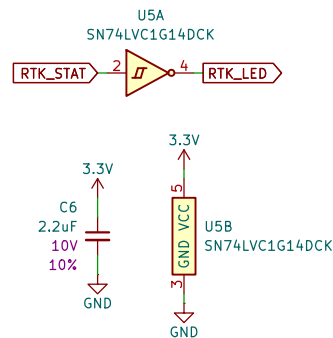
Option: IMU – IM19



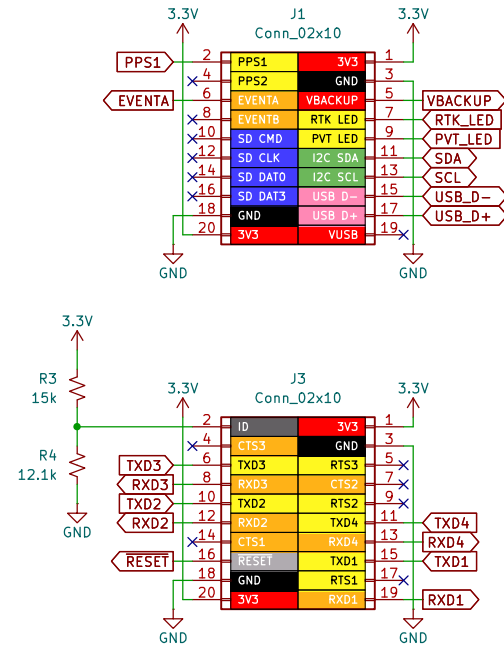
Antenna



RTK STAT Inversion 74LVC1G14



Connectors



Legend: GNSS Flex

Power +	GNSS Out
Power GND	GNSS In
PC	microSD
USB	ID (0-3.3V)
RMII	Aux

Microstrip (Coplanar Single Ended) 50Ω Calculation:
<https://lcpb.com/pcb-impedance-calculator>
 Board thickness: 1.6mm, Layers: 4, Er: 4.4 (7628 Prepreg)
 Dielectric Thickness (Layer 1 to 2): 8.28mil/0.2104mm (JLC7628)
 Copper Thickness (1oz outer / 0.5oz inner): 1.38mil/0.035mm
 Polygon Isolation: 7mil/0.1778mm
 Trace Width: 11.1mil/0.2819mm

USB Track Impedance: Coplanar Differential Pair @ 90Ω
<https://lcpb.com/pcb-impedance-calculator>
 Board thickness: 1.6mm, Layers: 4, Er: 4.4 (7628 Prepreg)
 Dielectric Thickness (Layer 1 to 2): 8.28mil/0.2104mm (JLC7628)
 Copper Thickness (1oz outer / 0.5oz inner): 1.38mil/0.035mm
 Polygon Isolation: 7mil/0.1778mm
 Trace Spacing: 9.07mil/0.2304mm
 Trace Width: 10.93mil/0.2776mm
 Trace center-to-center: 20.0mil/0.508mm



SPARKPNT



Designed by: P.C.

SparkFun Electronics

Sheet: /

File: ZED-X20P_Tilt.kicad_sch

Title: SparkFun GNSS Flex Module – ZED-X20P & IM19

Size: USLetter Date: 2025-05-15

KiCad E.D.A. 9.0.1

Rev: v10

Id: 1/1